ZARISKI MAIN THEOREM BY FINITE SECTION LIFTING AND THE FEASIBILITY OF A UNIT FORMALISATION

Scope and status. This note explains a projective proof of the version of Zariski Main Theorem used on page 1 of *Local structure of étale algebras* (`zmt2-revised.pdf`). It first gives a classical proof using formal functions, then isolates a finite constructive replacement. The latter extends the method of `finite-lifting.tex` and the two-equation ACL development. This document is a mathematical proof and a proposed formalisation programme, not a new Unit-checked theorem. The assessment below is based on the existing source interfaces and README files; no new Unit declarations were written or checked for this note.

Method of simplification. The first task is to work backwards from the output certificate, not to formalise the most general theorem appearing in the classical proof. Section 2 carries out this reduction. In particular, it removes the need for a constructive theorem saying that the entire ring of global functions is finite. The library programme below should be read with this smaller target in mind. Further analysis in Section 7 reduces the remaining input to finite top-degree coordinates from a bounded partial resolution. A relation among those coordinates is enough; even the kernel of the associated finite matrix need not be generated.

1. The exact statement needed

Theorem 1.1. *Let R be a commutative ring, $I \subseteq R$ an ideal, and $E = R[x_1, \dots, x_r]$ a finitely presented R-algebra. Suppose that E/IE is finite as an R/I-module. There exist a finite R-subalgebra $B \subseteq E$ and $s \in B \cap (1 + IE)$ such that the inclusion induces an isomorphism*

$$B[1/s] \xrightarrow{\sim} E[1/s].$$

Here and throughout, "finite" means finitely generated as a module; it does not mean free. A sufficient output certificate consists of $s \in 1 + IE$ and monic equations over R for

$$s, \quad sx_1, \dots, sx_r.$$

Indeed, $B = R[s, sx_1, \dots, sx_r]$ is then finite, and $x_i = (sx_i)/s$ proves the localisation assertion.

For the application to `zmt2-revised.pdf`, take $I = \mathfrak{m}$, where R is residually discrete local. The preceding finite-fibre argument in that note supplies that $E/\mathfrak{m}E$ is finite. Every prime $\mathfrak{q}$ containing $\mathfrak{m}E$ avoids $s \in 1 + \mathfrak{m}E$. This supplies exactly the initial finite subalgebra used there. No flatness, formal smoothness, or separability is used in Theorem 1.1 itself.

2. Working backwards: a smaller certificate

The finite algebra in Theorem 1.1 is an output, but we need not construct its multiplication table first. Nor do we need monic equations for arbitrary global functions. The following smaller certificate is sufficient, over any commutative base ring.

Lemma 2.1 (One finite stable module)**.** *Let $V \subseteq E$ be a finitely generated R-submodule containing 1. Suppose multiplication by each of $s, sx_1, \dots, sx_r$ preserves V. Then all these elements are integral over R.*

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Proof. Choose generators $v_1, \dots, v_m$ of V. For any one of the displayed elements t, write $tv_j = \sum_k a_{kj} v_k$, with $a_{kj} \in R$. The characteristic polynomial of (a_{kj}) is monic and annihilates every v_j by the adjugate identity. It annihilates $1 \in V$, hence vanishes at t in E. Independence of the v_j is not required. The matrices can be different for the different t. □

Let F_dE be the R-span of monomials of total degree at most d in $x_1, \dots, x_r$. It has an explicit finite generating list.

Lemma 2.2 (Degree-lowering certificate)**.** *Suppose that $s \in 1 + IE$ and, for each i, there are $N_i \geq 1$ and $q_i \in R[T_1, \dots, T_r]$ with*

$$sx_i^{N_i} = q_i(x), \qquad \deg q_i \leq N_i - 1. \tag{DL}$$

Then Theorem 1.1 holds for these data.

Proof. Choose a polynomial representing s, with degree bound $D \geq 0$, and put $d = D + \sum_i (N_i - 1)$. Relation (DL) gives

$$sF_{d+1}E \subseteq F_dE.$$

For a monomial x^α of degree at most $d+1$, if some $\alpha_i \geq N_i$, replacement by (DL) gives $sx^\alpha \in F_{|\alpha|-1}E \subseteq F_dE$. Otherwise $|\alpha| \leq \sum_i (N_i - 1)$, and the chosen polynomial for s gives $sx^\alpha \in F_dE$ directly. Thus F_dE contains 1 and is stable under s and all sx_i. Lemma 2.1 proves the assertion. □

Lemma 2.3 (Only chart denominators are needed)**.** *Let $W \subseteq \mathbf{P}^r_R$ have affine chart $\operatorname{Spec} E = D_+(Z)$, and let $h \in H^0(W, \mathcal{O}_W(-1))$. Set $s = (Zh)|_{\operatorname{Spec} E}$. Then s satisfies the degree-lowering relations (DL).*

Proof. On $D_+(X_i)$ represent h as $Q_i/X_i^{k_i}$ with $k_i \geq 1$ and Q_i homogeneous of degree $k_i - 1$ in the homogeneous coordinate ring. Choose a homogeneous polynomial representative. Agreement with $h = s/Z$ on $D_+(ZX_i)$ gives, after setting $Z = 1$,

$$sx_i^{k_i} = Q_i(1, x) \quad \text{in } E[1/x_i].$$

Clear this equality: for some $l_i \geq 0$,

$$sx_i^{k_i + l_i} = x_i^{l_i} Q_i(1, x) \quad \text{in } E.$$

Take $N_i = k_i + l_i$ and $q_i(T) = T_i^{l_i} Q_i(1, T)$, of degree at most $N_i - 1$. This uses only localisation equalities and degrees. □

Thus a lifted section gives the final certificate without computing the whole ring $H^0(W, \mathcal{O}_W)$, its generators, or its products. In fact the extraction needs only agreement of h on $D_+(Z)$ with each $D_+(X_i)$; agreement between the other charts is not used. Constructing a genuine global section remains one way to obtain this weaker datum. We do not claim here that a lifting theorem for the weaker overlap conditions has already been established.

3. The projective proof over a noetherian ring

For this section assume that R is noetherian. Choose a presentation

$$E = R[T_1, \dots, T_r]/(f_1, \dots, f_t).$$

Homogenise the f_j in $Z, X_1, \dots, X_r$, and let $W \subseteq \mathbf{P}^r_R$ be their common zero scheme. Then W is projective and

$$X := W \cap D_+(Z) = \operatorname{Spec} E, \qquad x_i = X_i/Z.$$

One need not compute the saturated ideal of the scheme-theoretic closure: this projective scheme, possibly with extra components at infinity, is sufficient. Put $\mathcal{F} = \mathcal{O}_W(-1)$.

Lemma 3.1. *For every $n \geq 1$, E/I^nE is finite over R/I^n.*

Proof. The image of each x_i in the finite algebra E/IE satisfies a monic polynomial. Lift it to $P_i(T) \in R[T]$. Then $P_i(x_i) \in IE$, so $P_i(x_i)^n \in I^nE$. The polynomial $P_i(T)^n$ is monic. Reducing powers of each of the finitely many generators by these monic equations gives a finite spanning family for E/I^nE. This argument also works without noetherianity. □

Write $W_n = W \times_R R/I^n$ and $X_n = \operatorname{Spec}(E/I^nE)$.

Lemma 3.2. *The open subscheme $X_n \subseteq W_n$ is also closed. There is a canonical section*

$$h_n \in H^0(W, \mathcal{F}/I^n\mathcal{F})$$

equal to Z^{-1} on X_n and zero on its complement. These sections are compatible as n varies.

Proof. By Lemma 3.1, X_n is finite, hence proper, over R/I^n. Since W_n is separated over that base, the map $X_n \to W_n$ is proper (use its closed graph). Being also an open immersion, it identifies X_n with an open-and-closed subscheme. The coordinate Z trivialises $\mathcal{O}(1)$ on X_n, so Z^{-1} is a section of $\mathcal{F}$ there. Glue it with zero across the disjoint complement. Both the decomposition and this section commute with reduction from $n+1$ to n. □

This is the general replacement for the particular almost-section $(0, 0, Z^{-1})$ used for two residually linear equations. The finite fibre need not consist of a single rational point.

Proposition 3.3. *There exists $h \in H^0(W, \mathcal{F})$ whose reduction modulo I is h_1.*

Proof. Set $M = H^0(W, \mathcal{F})$, a finite R-module by projective cohomological finiteness. The theorem on formal functions gives

$$\widehat{M} \;\cong\; \varprojlim_n H^0(W, \mathcal{F}/I^n\mathcal{F}).$$

Thus the h_n determine $\widehat{h} \in \widehat{M}$. For a finite module over a noetherian ring, $M/IM \cong \widehat{M}/I\widehat{M}$. Choose $h \in M$ representing the reduction of $\widehat{h}$. Naturality of the comparison maps gives the asserted reduction. □

Remark 3.4. We do not assert that $\widehat{h}$ comes from M. In general R is not complete, and that assertion would be false. Only its reduction modulo I must come from an actual section. Formal functions here is the comparison theorem for a proper morphism, a coherent sheaf, and an ideal of a noetherian base; see [4], Theorem 30.20.5.

Conclusion of Theorem 1.1 over noetherian R. Set $s = (Zh)|_X$. Its reduction is 1, because h reduces to Z^{-1} on X_1. Lemma 2.3 gives degree-lowering relations, and Lemma 2.2 produces monic equations for $s, sx_1, \ldots, sx_r$. Take $B = R[s, sx_1, \ldots, sx_r]$. It is finite and $B[1/s] = E[1/s]$. This proves exactly what is needed, without a separate finiteness theorem for $H^0(W, \mathcal{O}_W)$. □

4. A finite replacement for formal functions

Let $C^\bullet$ be the finite standard-chart Čech complex for $\mathcal{F}$ on W. Its terms need not be finite R-modules. Sections of the quotient on each affine intersection lift to sections of $\mathcal{F}$; consequently h_n can be represented by $z_n \in C^0$ with

$$dz_n \in I^nC^1, \qquad z_n - z_1 \in IC^0.$$

The second assertion follows from compatibility modulo I. Localisation is exact, so the subcomplex of sections of $I^n\mathcal{F}$ is precisely $I^nC^\bullet$. None of these assertions assumes that $\mathcal{F}$ is flat over R.

Proposition 4.1 (Finite lifting)**.** *Suppose the graded module*

$$\mathcal{H} = \bigoplus_{n\geq 0} H^1(I^n C^\bullet)T^n$$

over $\mathcal{R} = \bigoplus_{n\geq 0} I^n T^n$ *has the following constructive noetherian property: a sequence of homogeneous elements has a term which is a finite linear combination of its predecessors, with a certificate. Then there is* $z \in C^0$ *with* $dz = 0$ *and* $z - z_1 \in IC^0$*.*

Proof. For $n \geq 1$ set $\beta_n = [dz_n]T^n \in \mathcal{H}_n$. A dependence $\beta_N = \sum_{1\leq i<N} q_i\beta_i$, after taking degree N, gives coefficients $r_i \in I^{N-i}$ and $v \in I^N C^0$ with

$$dz_N - \sum_{1\leq i<N} r_i dz_i = dv.$$

Then

$$z = z_N - \sum_{1\leq i<N} r_i z_i - v$$

is closed. Since every r_i is in I, it has the desired reduction. The case $N = 1$ has an empty sum. Only finitely many approximations and one finite relation occur in the certificate. □

Classically, $\mathcal{H}$ is finite over the noetherian Rees algebra by projective cohomological finiteness applied over that algebra; this is also Lemma 30.20.1 of [4]. Constructively, one must produce the generators and the noetherian dependence mechanism. These are explicit obligations, rather than an appeal to completion. This proposition is the argument of `finite-lifting.tex`.

Under propositional truncation, a family of merely existing lifts must not be replaced by a chosen infinite sequence. Instead, given a coded noetherian bar, traverse its well-founded tree: at each node use one truncated lift and code, recurse into the proposition-valued goal, and stop at a certified dependence. The existing `GradedDependence` development is a model for this discipline; the generic construction above is not already its exported interface.

Relation to Artin–Rees and to flatness. Artin–Rees controls the induced filtration on a submodule:

$$N \cap I^{m+c}R^a \subseteq I^m N.$$

Our existing two-equation proof uses this to express a small removable obstruction with small coefficients on fixed removable obstructions. Their primitives then give a small correction of a cochain. Proposition 4.1 packages the analogous control in a graded module over the Rees algebra. One can pursue either implementation; using the graded proposition does not require first formalising the entire theorem on formal functions.

The flatness argument uses a different consequence of Artin–Rees, $\mathrm{Ann}(f) \cap f^N E = 0$ for sufficiently large N. Finite Taylor lifts place an annihilated element deep enough in this filtration to kill it. This explains the common algebraic mechanism, but flatness is not an input to the present ZMT proof. In particular, the Rees sheaf must use the images $I^n\mathcal{F}$, not replace them by $I^n \otimes_R \mathcal{F}$ without justification.

5. Reduction to an arbitrary base ring

Return to arbitrary R. From finiteness of E/IE, choose monic $P_i \in R[T]$ and finite witnesses

$$P_i(x_i) = \sum_{j=1}^{u} a_j H_{ij}(x), \qquad a_j \in I.$$

A single finite family a_j suffices for all i. Choose polynomial witnesses for these identities in the presentation of E; thus each difference is expressed as a combination of the f_l.

Introduce finitely many variables over $\mathbb{Z}$ for all coefficients in the presentation, the P_i, the H_{ij}, the a_j, and the identity witnesses. Impose the finitely many coefficient equalities which make the displayed identities hold. This constructs a finitely presented $\mathbb{Z}$-algebra R_0, an ideal $I_0 = (a_{1,0}, \dots, a_{u,0})$, and a finitely presented algebra E_0 with a map to the original data. Monicity is preserved by choosing the leading coefficients to be 1. The equations imply that E_0/I_0E_0 is finite, by reduction of powers.

Apply the noetherian-base result to (R_0, I_0, E_0). Transport its finite certificate to E. More explicitly, if $B_0 \subseteq E_0$ is finite, take the image of $R \otimes_{R_0} B_0$ in E. This is finite, contains s, sx_i, and has $s \in 1 + IE$. The map $R_0 \to R$ need not be injective. One must not instead assume that its image is a coherent ring or has a finitely generated kernel. This reduction does not require I itself to be finitely generated.

6. What the existing Unit developments provide

The following inventory records interfaces inspected for this note, not a fresh validation run of those developments. Paths are relative to `unit/sag/`.

- `artin-rees/CFN`, `ZCFN`, `HilbertCFN`, `QuotCFN`: coherent, strongly discrete, constructively noetherian rings, with the integer, polynomial and finite-quotient instances. These are suitable for the R_0 of Section 5.
- `artin-rees/FreeCoherent`, `FreeDisc`, `Saturation`, `DegreeCor`, `Schreyer`: finite syzygies, bounded-degree intersections and polynomial coherence. These are ingredients for general elimination and graded presentations; they are not already a projective cohomology API.
- `acl-zmt/Laurent3`, `Cech3`, `CechContraction`, `CechMonomials`, `CechH1Finite`: explicit three-coordinate Laurent and Čech calculations. Their interfaces concern $\mathbf{P}^2$, not arbitrary $\mathbf{P}^r$.
- `acl-zmt/FiniteCorrection`, `GradedDependence`, `ReesLiftingCong`: the finite correction and graded dependence mechanisms, in interfaces using two ideal generators and particular finite-coordinate obstruction data.
- `artin-rees/ArtinRees`, `PowSpan`, `ACLRelations`, `ACLCorrection`: the alternative correction by Artin–Rees for $I = (a, b)$, with the two-equation obstruction matrix. The general number of ideal generators is still an extension.
- `acl-zmt/IntegralSections` and the exterior-algebra determinant files provide models for the passage from finite sections to monic equations. `ACLCertificateGeneral` assembles the current two-equation case, with its stated gcd-domain and CFN hypotheses.

The directory `sag/projective` constructs particular projective line examples and coordinate calculations, with additional SAG hypotheses in some results. Its name does not indicate a library of general projective cohomological finiteness. The proposed proof can stay with algebraic localisations and finite Čech complexes and need not import those extra hypotheses.

7. Further reductions of the remaining lifting problem

7.1. **The one-variable case has a direct certificate.** Suppose $E = R[x]$. Lift a residual monic relation of degree $d \geq 1$ and its I-membership witnesses to a polynomial relation

$$f(x) = 0, \qquad f(T) \equiv P(T) \pmod{IR[T]}, \qquad P \text{ monic}, \quad \deg P = d.$$

Writing $f(T) = \sum_{j=0}^{m} c_j T^j$ with $m \geq d$ (padding with zero coefficients if necessary), put

$$s = c_d + c_{d+1}x + \cdots + c_m x^{m-d}.$$

Then $s \in 1 + IE$ and

$$sx^d = -\sum_{j<d} c_j x^j.$$

This is exactly (DL); Lemma 2.2 finishes without any noetherianity, formal lifting, Artin–Rees, or cohomology. If a residual relation has degree zero, enlarge it to positive degree. This is a useful base case and an explicit test of what the projective machinery must accomplish in more variables.

7.2. **The apparent weaker-overlap shortcut has a cost.** Section 2 observes that extraction only uses overlaps with $D_+(Z)$. It does not follow that one should simply discard the other overlaps from the lifting complex. For example, on $\mathbf{P}^2_k$, retain just the edges XZ and YZ. In degree -1, the edge cochains

$$(X^{-n}Y^{2n-1}Z^{-n},\ 0), \qquad n \geq 1,$$

give linearly independent classes modulo chart coboundaries. A chart X cochain has nonnegative Z exponent; a chart Z cochain has nonnegative X exponent; neither can contribute the displayed monomial on XZ. The chart Y does not contribute on that edge. Thus this shortened complex has infinite-dimensional cokernel, although the actual $H^1(\mathbf{P}^2_k, \mathcal{O}(-1))$ is zero. Keeping only the conditions used at the end can destroy the finiteness that makes lifting possible. A different argument for the weaker conditions would have to address this explicitly.

7.3. **Replace general cohomology finiteness by one finite matrix.** Here is a more effective reduction. Let D be CFN, $S = D[X_0, \ldots, X_r]$, and $M = S/J$, with J a finitely generated homogeneous ideal. Write all the following modules with their twist by -1 already included. Choose a partial homogeneous free resolution

$$F_{r-1} \xrightarrow{\partial_{r-1}} F_{r-2} \longrightarrow \cdots \longrightarrow F_1 \longrightarrow F_0 = S(-1) \longrightarrow M(-1) \longrightarrow 0. \tag{PR}$$

It must be exact at $F_0, \ldots, F_{r-2}$; no condition on the kernel at the left end is imposed. Each F_j is a finite sum of shifts of S. For $r = 2$, F_1 is just a generating list for J, shifted. For larger r, only $r - 2$ additional rounds of homogeneous syzygies are required after that generating list.

Proposition 7.1 (A finite source for the needed classes)**.** *For $r \geq 2$, form the finite matrix*

$$A : H^r(\widetilde{F}_{r-1}) \longrightarrow H^r(\widetilde{F}_{r-2})$$

induced by ∂_{r-1}. There is a surjection

$$\ker A \ \twoheadrightarrow\ H^1(\widetilde{M}(-1)).$$

The source of A and its target are finite free D-modules, with the usual all-negative-exponent monomial bases for each shift. Consequently coherence of D supplies a finite generating family for the required H^1 classes.

Proof. Put $K_0 = M(-1)$ and use the exact sequences $0 \to K_{j+1} \to F_j \to K_j \to 0$ through $j = r - 2$, with $F_{r-1} \twoheadrightarrow K_{r-1}$. Intermediate cohomology of every shifted free module vanishes. Also $H^q(\mathcal{O}(-1)) = 0$ for all q, so the first connecting map gives $H^1(K_0) \cong H^2(K_1)$. Continuing the connecting maps gives, at the last step, an identification of $H^1(K_0)$ with

$$\ker\big(H^r(K_{r-1}) \to H^r(F_{r-2})\big).$$

For $r = 2$ this is the first step itself. Since there is no cohomology above degree r, the map $H^r(F_{r-1}) \to H^r(K_{r-1})$ is surjective. The preimage of the displayed kernel is precisely $\ker A$. This proves the assertion.

The maps can be realised as finite Čech zigzags: choose Laurent representatives in top degree, use $Av = 0$ to remove the top obstruction by the free-module monomial contraction, and descend through (PR). Conversely, lift a given 1-cocycle through (PR) and apply these contractions to obtain a vector in $\ker A$. A formal proof must implement both the zigzag and its equality witnesses; the long exact sequence argument identifies exactly what to implement. □

For $r = 2$, the target is $H^2(\mathcal{O}(-1)) = 0$: every top-degree vector is admissible. This recovers the special shape of `CechH1Finite`. For $r = 3$, one syzygy module and one finite kernel calculation suffice. For $r = 1$, $H^1(\widetilde{M}(-1)) = 0$ directly, since it is a quotient of $H^1(\mathcal{O}(-1))$ and the complex has no degree 2. The direct polynomial argument above is preferable for this base case.

Remark 7.2 (A concrete first test in four coordinates). Take $J = (X_0^3, X_1^3)$ in $D[X_0, X_1, X_2, X_3]$. The pair of variable powers is regular, so its two-step Koszul resolution gives

$$F_2 = \mathcal{O}(-7), \qquad F_1 = \mathcal{O}(-4)^2, \qquad F_0 = \mathcal{O}(-1), \qquad \partial_2 = (-X_1^3, X_0^3).$$

Here the notation denotes the associated sheaves of the shifted free modules. The source of A has the 20 monomials $X_0^{-a_0} \cdots X_3^{-a_3}$ with $a_i \geq 1$ and $\sum a_i = 7$; the target has one all-negative monomial in each of its two summands. Its first row selects, with a minus sign, $(a_0, a_1, a_2, a_3) = (1, 4, 1, 1)$; its second selects $(4, 1, 1, 1)$. All other columns vanish. Thus its kernel is freely generated by the remaining 18 monomials, over any D. The zigzag must turn these into 18 generating 1-cocycles. This tests a genuinely nonzero top matrix without first requiring an arbitrary syzygy algorithm. It is a test of the cohomology reduction, not an example of a finite affine fibre. No Unit check of this example is claimed.

7.4. **A further shortcut: do not generate the finite kernel.** The last sentence of Proposition 7.1 still proves more than we need. Let $V = H^r(\widetilde{F}_{r-1})$, a fixed finite free D-module. For each particular obstruction class β_n, the surjectivity argument gives a vector $v_n \in V$ with $Av_n = 0$ and image β_n. Any D-linear relation among these chosen vectors is a relation among their images. Noetherian dependence can be applied in V itself. There is no need to find generators of $\ker A$, or even to construct A as a separate matrix interface. Its kernel condition describes the domain of the connecting map; the given zigzag certificates already supply that condition.

The useful formal interface is consequently smaller than Proposition 7.1: send a given 1-cocycle to finite top-degree coordinates with a zigzag certificate, and prove that a relation among finitely many such coordinates produces a primitive for the same relation among their cocycles. The coordinate assignment need not be canonical or additive. What must be additive is the connecting construction on the certified top cycles. Constructive noetherianity is used only in the ambient finite free module.

Lemma 7.3 (The minimal relation-certificate interface)**.** *Let D be CFN, Z a D-module, $B \subseteq Z$ a submodule, and $V = D^v$. Suppose a relation $\Gamma \subseteq Z \times V$ contains $(0, 0)$ and is closed under addition, negation and D-scalars. Suppose also*

$$\Gamma(z, 0) \implies z \in B.$$

Let $b_1, b_2, \ldots$ be given elements of Z, each merely admitting $p_n \in V$ with $\Gamma(b_n, p_n)$. Then there merely exist $N \geq 1$ and $Q_i \in D$ for $1 \leq i < N$ with

$$b_N - \sum_{i<N} Q_i b_i \in B.$$

Proof. Use noetherian free-list induction in V, retaining in a finite prefix the chosen p_i and their certificates. At the next index, eliminate the truncated existence of one further p_n into the proposition-valued goal. If it is in the span of its predecessors, extract a finite dependence; closure of Γ gives $\Gamma(b_N - \sum Q_i b_i, 0)$, hence the conclusion. Otherwise extend the free list and recurse. This uses the same induction as `SeqDependence`, with the choices inside the induction rather than a chosen infinite sequence. The weighted homogeneous version retains the degree witnesses and takes the appropriate homogeneous parts of the coefficients in the resulting finite relation. □

In the application, Z consists of the relevant graded cocycles, B consists of coboundaries, and Γ is the existence of a finite zigzag certificate. The lemma only needs module predicates and

finite lists, not quotients of Z or a category of complexes. It makes the remaining implementation obligation precise: construct the particular zigzag relation and prove its closure and zero-vector property. No further mathematical shortcut past this obligation is established in this note.

7.5. **Use polynomial coordinates instead of building a Rees API.** Apply the preceding reduction to the bigraded presentation of $N = \bigoplus_n I_0^n(S/J)T^n$ described below, over $D = R_0[U_1, \dots, U_u]$. Keep projective degree and Rees degree separate. The top monomial lists are finite over D and have fixed Rees shifts $e_1, \dots, e_v$. The raising zigzag for $[dz_n]T^n$ gives top-degree vectors $p_n = (p_{n,j}(U))_j$, with $\deg_U p_{n,j} = n - e_j$ (zero when this is negative), together with zigzag witnesses. We only need a dependence among these vectors in the finite free D-module D^v. Constructive noetherian dependence there gives

$$p_N = \sum_{1 \le i < N} Q_i(U) p_i.$$

Taking weighted degree N, with the jth coordinate shifted by e_j, replaces Q_i by its homogeneous part of degree $N - i$. Substitute $U_j = a_j$: its value r_i belongs to I_0^{N-i}. The relation-reflection property of the zigzag gives precisely

$$dz_N - \sum_{i<N} r_i dz_i = dv, \qquad v \in I_0^N C^0,$$

and finite correction finishes. Terms of negative coordinate degree are treated as zero, rather than assigned fictitious polynomials.

Thus no general cohomology quotient construction or standalone Rees ring library is needed for the dependence step. One needs a bigraded polynomial presentation, a bounded partial resolution, finite top-coordinate lists, and the certificate that their relations give coboundaries. Generators for the cohomology or the top matrix kernel are unnecessary. The noetherian-tree treatment of truncated choices remains necessary. This is closely matched to the existing finite-coordinate `GradedDependence` mechanism.

8. The new work and its feasibility

8.1. **Finite-fibre sections in an algebraic chart model.** Lemma 3.1 and the final certificate are short algebraic tasks. For Lemma 3.2, one can avoid formalising the general properness argument. Here is an algebraic construction of its chart idempotents.

Work modulo I^n on the chart $D_+(X_i)$, with coordinate ring A_i, and put $u = Z/X_i$. On its intersection with $D_+(Z)$ we have $x_i = u^{-1}$. A monic relation of degree d for x_i gives

$$H(u) = 1 + c_1 u + \cdots + c_d u^d = 0 \quad \text{in } A_i[1/u].$$

Equality in a localisation supplies $L \ge 1$ with $u^L H(u) = 0$ in A_i. Since $H(0) = 1$, there are polynomials $A(U), B(U)$ with

$$A(u)u^L + B(u)H(u) = 1.$$

They can be obtained by the finite geometric-series inverse of H modulo U^L. The element $e = A(u)u^L$ is idempotent, and $D(e) = D(u)$. On the e-part, u has inverse $t = eA(u)u^{L-1}$, extended by zero on the complement; indeed $ut = e$ and $et = t$. The local section is tX_i^{-1}, whereas on $D_+(Z)$ it is Z^{-1}. These agree on overlaps because both are Z^{-1} on the affine part and zero on its complementary idempotent part. The same characterisation proves compatibility under reduction.

Thus the finite-fibre section construction has explicit polynomial certificates. The new formal work is to implement these formulas, their chart changes and their overlap identities. The existing `FixedCochain` treats only the residual-point special case.

8.2. **Any finite number of projective coordinates.** Define Laurent polynomials on $\mathbb{Z}^{r+1}$ with finite support, partial localisation support conditions, and the alternating complex indexed by nonempty subsets of the coordinate set. Generalise the monomial contraction: the negative coordinates of a monomial determine its possible chart components. This gives vanishing in intermediate degrees for $\mathcal{O}(k)$ and a finite monomial basis in top degree, as well as the degree-zero calculation. Coefficients need not have decidable equality for this contraction; the exponent comparisons are decidable. This is a substantial generalisation of the three-variable code, with a clear combinatorial proof.

8.3. **Finite presentations of the cohomology.** For our end result, the finite top-coordinate certificates of Section 7 suffice. A general cohomology-presentation interface would do extra work. No H^0 calculation is needed for the extraction of monic equations: Section 2 already supplies a smaller argument.

For a finitely presented graded module over $D[X_0, \dots, X_r]$, with D CFN, construct successive homogeneous syzygies and finite sums of shifted free modules mapping onto them. Proposition 7.1 explains why terms through F_{r-1} suffice. Its further reduction avoids computing the finite kernel: lift only the given obstruction classes to top-coordinate vectors, and use dependence in their ambient finite free module. Coherence is still used to construct the successive homogeneous syzygies over the polynomial ring, not to present the resulting cohomology.

This is an implementation strategy, not an existing Unit theorem. The truncation bound, exactness, lifts and comparison maps must be proved. In particular, one must not assume that an arbitrary module over a singular base has a terminating finite free resolution. No general Hilbert syzygy theorem is required for a bounded initial resolution. The convenient two-equation gcd syzygy calculation is replaced here by iterated homogeneous coherence.

8.4. **Presenting the Rees module without a flatness assumption.** Let $S = R_0[X_0, \dots, X_r]$, J the chosen homogeneous ideal, $M = S/J$, and $I_0 = (a_1, \dots, a_u)$. The relevant graded algebra is

$$N = \bigoplus_{n \geq 0} I_0^n M\, T^n \subseteq M[T].$$

Set $D = R_0[U_1, \dots, U_u]$. The map

$$D[X_0, \dots, X_r] \longrightarrow M[T], \qquad U_j \longmapsto a_j T$$

has image N. Its kernel is the elimination ideal

$$\big(J,\ U_1 - a_1 T, \dots, U_u - a_u T\big) R_0[U, X, T] \ \cap\ R_0[U, X].$$

It is not in general enough to impose J and the familiar pairwise Rees relations: M was not assumed flat over R_0.

There is a useful potential reuse here. Treat T as the final polynomial variable over the CFN ring $R_0[U, X]$. The degree-zero intersection case of `DegreeCor` supplies generators of precisely this elimination ideal, with membership witnesses. Thus a new multivariate Gröbner-basis implementation need not be a prerequisite. One still has to implement the bridge to that interface, extract homogeneous components for the projective and Rees gradings, and prove the kernel/image presentation. This proposal has not been instantiated in Unit for the present ideal.

Applying the finite-coordinate construction over D to this presentation controls the required classes in

$$\bigoplus_n H^1(W, I_0^n \mathcal{O}_W(-1)).$$

One must prove that localisation, the finite chart complex and extraction of projective degree commute with the direct sum. Keep polynomial coordinates and use dependence in D^v, as in Section 7. There is no need to construct a separate formal Rees-ring interface or generate the entire

graded cohomology module. The two gradings and their comparison maps remain a significant source of representation work.

8.5. **Assessment and suggested milestones.** The proof is feasible in the foundations and algebraic style already used by Unit: finite supports, quotients, images, propositional truncations and coded noetherian trees. There is no identified need for countable choice, excluded middle, flatness, or a full formal-functions library. Nevertheless it is a substantial library extension, not a short instantiation of the existing ACL theorem. The sharper target is the free-module monomial contraction and the finite-coordinate zigzag explained by Proposition 7.1, together with their bigraded presentation bridges. A general projective cohomology library is optional. The polynomial-coordinate route also avoids generalising Artin–Rees itself to arbitrarily many ideal generators. Only finite ideal powers and elementary correction formulas need that generalisation. The finite-fibre section construction remains a separate task with explicit chart formulas.

A useful order of work, revised by working backwards, is:

(1) Formalise Lemmas 2.1–2.3: extract the certificate using bounded-degree monomials and chart denominators. Then formalise the finite-presentation descent of Section 5. Neither task needs cohomological finiteness.
(2) Generalise ideal powers and finite correction to a finite list of generators. Use noetherian dependence over D to produce the weighted polynomial-coordinate relation. State the lifting result conditionally on its explicit finite-coordinate reduction data.
(3) Implement finite-fibre sections and the Laurent/Čech contraction for any fixed finite number of coordinates. Check the new construction against the existing $\mathbf{P}^2$ formulas.
(4) Implement the raising Čech zigzag and its relation-reflection certificate, initially for a supplied partial homogeneous resolution. Then construct the terms through F_{r-1} using CFN coherence. Do not compute a top matrix kernel or cohomology generators unless another application needs them.
(5) Instantiate elimination for N, verify the Rees/Čech comparison, and discharge the graded finiteness hypothesis. Assemble the arbitrary-base theorem and the instance $I = \mathfrak{m}$.

Each milestone should export a theorem with its genuine parameters. A conditional lifting result is useful, but does not prove ZMT until the finiteness and finite-fibre inputs are constructed. The first decisive feasibility check beyond the existing $\mathbf{P}^2$ code is a $\mathbf{P}^3$ example: one homogeneous syzygy module, the 20 top monomials in the cubic example above, a raising zigzag, and the primitive obtained from a relation among its top-coordinate vectors. No reliable calendar estimate follows from the present inspection.

8.6. **Current stopping point and practical feasibility.** The simplifications above leave a concrete finite-algebra programme. We do not have a proved route which dispenses with the uniform finite-coordinate construction. Merely imposing fewer overlap conditions fails by the infinite-dimensional example above; merely applying noetherianity to the original chart modules is unavailable, since those modules are not finite over the base. These observations identify the remaining difficulty, not an impossibility theorem for all other approaches.

The existing determinant trick, polynomial coherence, saturation, finite free-module dependence and finite correction provide the algebraic ends of the proof. The principal new proof is the zigzag relation of Lemma 7.3, for general finite numbers of coordinates and a supplied partial resolution. Weighted homogeneous syzygies, the elimination presentation and the comparison with I^n-images then connect it to the application. Existing coherence is ungraded: extracting and tracking the two homogeneous gradings is new interface work, not an already verified automatic operation.

A prudent first implementation should therefore stop at two precise deliverables: the abstract relation-certificate lemma, including truncated choices; and the four-coordinate cubic example with its zigzag and relation-to-primitive theorem. Completing those would validate the new mechanism.

Completing only the abstract lemma would not validate projective lifting. The full theorem is now a realistic but substantial formalisation, with representation and comparison proofs as the main risks. There is no basis yet for calling it a small patch or assigning a reliable completion time. A machine-checked mathematical certificate is the target; an efficient executable ZMT algorithm would be a further requirement.

9. Checks which should accompany a formalisation

The output should include $s-1=\sum_j a_j r_j$ and explicit monic equations for s, sx_i, with polynomial ideal-membership witnesses in the presentation of E. Finite generation rather than freeness must be used throughout. Useful regression cases are:

- the existing two-variable ACL example, to recover the current certificate theorem;
- $E=R[t]/(t^2)$ and $I=0$, to ensure that a nonreduced finite fibre is allowed;
- $R=k[a]$, $I=(a)$, $E=R\times R[1/a]$, with the diagonal R-algebra structure: $B=R\times R$ and $s=(1,a)\in 1+IE$ work. The affine fibre is one component; a component invisible modulo I must not be lost before localisation;
- $E=R[1/a]$, $I=(a)$, whose special fibre is zero. A valid certificate may have $s=0$, since $IE=E$, giving zero localisations;
- a nonflat finite algebra over R, to detect accidental use of $I^n\otimes\mathcal{F}=I^n\mathcal{F}$.

The proposed new Unit files should follow the proof one lemma per declaration, with explicit structures and named comparison maps as in `seg/WRITING.md`. Validation must include fresh Unit checks of those files and their exact exported assumptions. Compiling this LaTeX note only validates the document, not the proposed formalisation.

References

[1] G. Horrocks, *Projective modules over an extension of a local ring*, Proc. London Math. Soc. (3) 14 (1964), 714–718; local file `phorrocks.pdf`. The section lifting in Theorem 1 is the motivation for this note.

[2] T. Coquand, *Local structure of étale algebras*, local file `/Users/coquand/COLL/zmt2-revised.pdf`, especially p. 1.

[3] M. Alonso, T. Coquand and H. Lombardi, *Revisiting Zariski Main Theorem from a constructive point of view*, Journal of Algebra 406 (2014), 46–68.

[4] The Stacks Project, Section 30.20, The theorem on formal functions, especially Lemma 30.20.1 and Theorem 30.20.5. The proofs above specify the particular consequence needed rather than claiming a new proof of the entire theorem.

[5] Internal notes in `unit/doc/`: the files `finite-lifting.tex` and `finite-cohomology.tex`; also `artin-rees.tex` and `artin-rees-acl-elementary.tex`. Their historical status paragraphs should be read with the current source and README files.